Name / Имя: _____ Class: _____ Date: _____

Laboratory Work No. 3 · Testing for organic molecules

Aim: find out which foods contain carbohydrates (starch), lipids and proteins.

Safety check · tick each box · Техника безопасности

✓	Rule
☐	I wear gloves and eye protection when reagents are used.
☐	I never taste the food samples.
☐	Only the teacher pours iodine solution and Biuret reagent.
☐	I tell the teacher about spills at once.
☐	I wash my hands after the lab.
☐	I told the teacher if I am allergic to egg or olives.

1 · Pre-lab questions · Вопросы до работы

a) Why do we need carbohydrates?

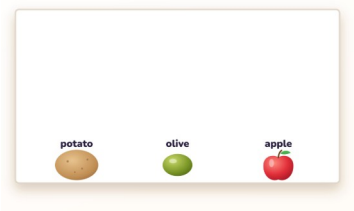
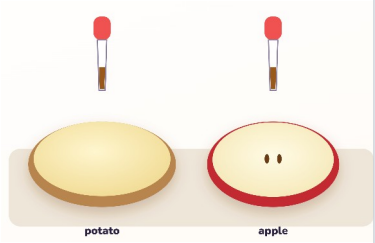
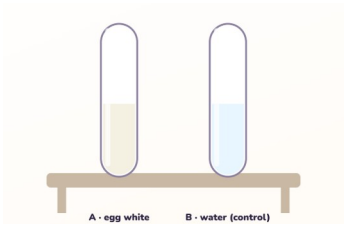
b) What is the role of proteins?

c) What is the importance of fats?

2 · Predict · Предскажи

Food	I think it contains… (carbohydrate / lipid / protein)	Because…
potato		
apple		
olive		
egg white		

3 · Procedure · Ход работы

Test	What to do	Result rule
A · Paper test (lipids) 	1. Squeeze and knead potato, olive and apple separately on white paper until wet. 2. Clean off the food. Let the paper dry for 10–15 minutes. 3. Hold the paper up to the light. Compare the traces.	If the food contains fat, the trace is **transparent**.
B · Iodine test (carbohydrates) 	1. The teacher cuts the potato and apple in half. 2. Add a few drops of iodine solution to each. 3. Write the colour you see.	With carbohydrate (starch), colour changes from **brown to blue or black**.
C · Biuret test (proteins) 	1. Put 2 ml raw egg white in a test tube. 2. The teacher adds 2 ml Biuret reagent. Mix. 3. Write the result. Compare with water.	With protein, colour changes from **blue to purple**.

Start test A first — the paper needs time to dry.

4 · Observations · Наблюдения

Write what **you** saw. Do not copy from the board.

Test	Food	Colour / trace I saw	Compound present? (yes / no)
Iodine	potato		
Iodine	apple		
Paper	potato		
Paper	olive		
Paper	apple		
Biuret	egg white		
Biuret	water (control)		

5 · Explain · Объясни

I observed _____. This suggests _____ because _____.

6 · Post-lab questions · Вопросы после работы

1. What foods contain carbohydrates?

2. What foods are rich in protein?

3. From what foods do we get fat?

4. Which result was hard to see? How could we make the test fairer?